```
import socket
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("0.0.0.0", 5000))
server.listen(1)
print("Server is waiting for connection...")
conn, addr = server.accept()
print("Connected to:", addr)
while True:
    data = conn.recv(1024).decode()
    if not data:
        break
    print("Client:", data)
    conn.send(data.encode())
conn.close()
server.close()
```

```
import socket
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server_ip = input("Enter server IP: ")
client.connect((server_ip, 5000))
while True:
    message = input("Client: ")
    client.send(message.encode())

    if message.lower() == "exit":
        break

    reply = client.recv(1024).decode()
    print("Server:", reply)

    if reply.lower() == "exit":
        break

client.close()
```

```
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("0.0.0.0", 5000))
server.listen(1)
print("Chat server waiting for connection...")
conn, addr = server.accept()
print("Connected to:", addr)
while True:
    message = conn.recv(1024).decode()

    if message.lower() == "exit":
        break
    print("Client:", message)
    reply = input("Server: ")
    conn.send(reply.encode())
    if reply.lower() == "exit":
        break
conn.close()
server.close()
```

Ex no.12

Nmap to discover live hosts

Aim

To discover live hosts in a network using Nmap by performing ARP, ICMP Scan, TCP Ping Scan and UDP Ping Scan on the Try hack me Platform

Procedure:

→ Log into the try hackme platform.
→ Open the Nmap room and start the target
→ Start the Attack box and open the terminal
→ Enter the target machine's IP address.
→ Perform ARP, ICMP, TCP and UDP scans using Nmap commands
→ Observe and record the live hosts discovered

Result:

Thus, live hosts in the network were discovered successfully.